

Machine Learning For Early Dynamic Prediction Of Functional Outcome After Stroke

Corresponding Author: Professor Emmanuel Carrera

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Summary

The study presents a series of non-linear machine learning models that were developed and validated to predict the mortality and functional status of patients suffering from ischemic stroke within three months of their original hospital admission. The models integrate a variety of clinical, physiological, and biological data collected within the first 72 hours of hospital admission. The authors leverage data from the Geneva University Hospital from 2018 to 2021 to derive the models and the MIMIC-III database to validate them externally. A comparison of the discriminatory performance of a transformer-based neural network architecture, a recurrent neural network architecture, and the XGBoost algorithm is illustrated in the manuscript. The Shapley Additive Plots (SHAP) were applied to identify the key predictors driving each patient's outcome.

Comments

The study illustrates the application of a series of well-established machine-learning techniques to two mid-term stroke patient outcomes after hospital admission. I would like to commend the authors for using two different patient cohorts from different countries to ensure the generalizability of the proposed models. An additional strength of the paper is the diverse types of data that are used to characterize a patient's condition and the incorporation of both time-variant and time-invariant information. Nevertheless, there are various areas of improvement for the manuscript that I outline below.

1. In its current form, the exposition of the manuscript does not meet the standards of the journal. I urge the authors to review the formatting and notation used across the manuscript to ensure that they conform to the international guidelines. In addition, the language used across the paper could be formalized, using more specialized vocabulary that is appropriate for a medical publication (see the second sentence of the Introduction).
2. One of my greatest concerns relates to the definition of the outcome of interest. Most of the literature focusing on three-month mortality after an acute ischemic stroke sets the timepoint of data collection at the time of hospital discharge. On the contrary, most researchers leveraging time-varying data throughout the hospital admission recognize in-hospital outcomes, such as in-hospital mortality and morbidities, length of stay, and hospital readmission, as the priority endpoints that affect their decision-making process. Why do the authors focus on the outcomes of functional status and mortality in a three-month horizon, leveraging up to 72 hours of information instead of at the time of discharge? As long as the patient remains at the hospital, it is inevitable that the patient's status will change, and thus, the specific prediction after 72 hours becomes irrelevant. According to the literature review, the median and average length of stay after an acute ischemic stroke is around seven to ten days (see [1-2]), which raises doubts regarding the clinical relevance of these models.
3. I would recommend to the authors to alter their focus, following one of the two proposed directions outlined below:
 - a. Predict short-term outcomes like 3-month mortality or functional status at the time of discharge using varying time horizons of observation for each patient depending on the length of stay for their index admission.
 - b. Predict in-hospital mortality and morbidity outcomes using the 72-hour time frame that they have already established in the manuscript by changing the dependent variables in the model.
4. Another issue with the selected outcomes is that there is limited value in a flexible model that is adaptive to hourly information. Given that the endpoint lies in a time horizon that is significantly distant compared to the input information, the ability of the model to incorporate a flexible timeframe is not reflected in its application. This observation reinforces the need to choose a different scope (as described in the previous comment).
5. The literature review is incomplete as there have been other approaches that use time-series ICU data to predict short-term stroke mortality with transformer-based architectures [3-4]. The articles referenced above were retrieved after a quick search in online libraries. I recommend a more thorough review of the existing medical literature, including pre-trained

models that use a transformer-based architecture that are tailored to the task of mortality prediction after an acute ischemic stroke, even with different types of data.

6. How did the authors filter the population to ensure that it involves only patients with acute ischemic stroke? As highlighted in [5], ICD codes are notoriously inaccurate in capturing the true clinical condition of the patient. The specific study found that more than 30% of the patients reported in insurance claims with acute ischemic stroke did not actually have the condition. I would suggest conducting an independent audit of a random sample of the patients included in the study with expert physicians who could verify the patients' diagnosis.

7. The study period involves some of the initial phases of the COVID-19 pandemic. Do the models account for whether a patient's stroke was provoked by COVID-19 or not? Can the authors provide any insights regarding the impact of COVID-19 on the mortality risk and functional status of patients who experience acute ischemic strokes?

8. There is a significant risk of cherry-picking the models when there is no bootstrapping with multiple random splits of the data between training/validation and testing sets in the derivation cohort. While the authors conduct bootstrapping the cross-validation procedure, it is important to apply an equivalent process for the training/testing set split and report confidence intervals around AUC, accuracy, specificity, etc.

9. It is important for the readers to understand better the hyperparameters and specific architectures used for the neural network models. What kind of structure did the authors use? Did it involve a pre-trained NN? How many observations were generally used per admission?

10. The fact that some patients were associated with multiple admissions might positively bias the performance of the model. I recommend including only the first admission per patient.

11. When delving into the results, the XGBoost model has equivalent and, in some cases, even better performance compared to the transformer-based model. The XGBoost model, though, does not have the ability to incorporate varying time-series information. How did the authors train this model? What was the structure of the feature vector that contained the time-series information? Why should clinicians prefer a more complex model that would be harder to implement in practice compared to the ensemble learner?

12. The authors could add a dedicated section in the Supplemental Material with more information regarding the mRS scale and the implications of each status.

References:

1. Chang, Ku-Chou, et al. "Prediction of length of stay of first-ever ischemic stroke." *Stroke* 33.11 (2002): 2670-2674.
2. Huang, Ying-Chih, et al. "The impact factors on the cost and length of stay among acute ischemic stroke." *Journal of Stroke and Cerebrovascular Diseases* 22.7 (2013): e152-e158.
3. Huang, Ruixuan, et al. "Stroke mortality prediction based on ensemble learning and the combination of structured and textual data." *Computers in Biology and Medicine* 155 (2023): 106176.
4. Antikainen, Emmi, et al. "Transformers for cardiac patient mortality risk prediction from heterogeneous electronic health records." *Scientific Reports* 13.1 (2023): 3517.
5. Ong, Charlene Jennifer, et al. "Machine learning and natural language processing methods to identify ischemic stroke, acuity and location from radiology reports." *PloS one* 15.6 (2020): e0234908.

Reviewer #2

(Remarks to the Author)

This manuscript addresses the prediction of functional outcome of stroke patients based on continuously acquired clinical data points. The methodology and experimental framework are well described, and the authors are doing an excellent job at introducing the landscape of stroke and the uncertainty associated with its treatment.

In particular, the evaluation of the framework is well designed as the models are trained on a large cohort of patients from the authors' institution and evaluated on a publicly available dataset with similar variables. This dramatically boosts the readers' interest as it is a good indication of the generalization power of the models introduced in the study.

While I did not notice significant weaknesses, a few concepts are a bit vague and should be more precisely described:

- Three machine learning methods were used and compared. While LSTM can take into account temporal sequences, it is not clear how the other two methods (XGboost and Transformer) were being used on continuous data. Was the input the 1-hour summary/average? How about for LSTM - was it a sliding window? If the data of each hour is inputted independently to the model, what patterns could be missing? Is there any temporal trend that could be captured with the current approach?
- ROC and PR curves are reported in figure 2 and represent the performance of the models on the MIMIC III dataset. What were the results on the training dataset after cross-validation?
- As mentioned by the authors, imaging findings were left out of the model. There is a great opportunity to evaluate them within the framework as it would provide an indication of their predictive power (even if only done on subset of patients from the local cohort). While using raw imaging is challenging, it seems very doable to incorporate findings from the neuro-radiology report such as volume of the core, penumbra, bleeding, and other commonly reported features. This would greatly improve the originality and the impact of the study.

Version 1:

Reviewer comments:

Reviewer #3

(Remarks to the Author)

The authors have satisfactorily addressed previous reviewer concerns. They have included additional clarifying details and context that improves the quality of the manuscript.

Reviewer #4

(Remarks to the Author)

Reviewer comments for rebuttal

Thanks for the opportunity in reviewing the original research work written by Klug et al, titled as 'Early dynamic prediction of functional outcome after stroke using machine learning'. This work has been submitted, recommended to be revised, and being reviewed again for its appropriateness and completeness for journal publication.

I found the article to be very intriguing since there are not a lot of literature suggesting the ML models' role in addressing functional outcome after acute stroke. However at the same time, ironically, the results authors presented also showed the characteristic 'prediction was predictable' tendency that comes with many ML models that sometimes nullify its value. performance of the Transformer (encoding) model was not drastically better than the XGBoost, and raised question on the absolute superiority of the sophisticated ML model with high-level inference capacity on this type of prediction. For example, functional outcome (authors used modified Rankin) could be all very predictable if NIHSS, pre-stroke disability, or IVT timing (which SHAP showed) were given to general neurologists in clinical setting without using ML models in their practice. I think the authors could probably construct a logistic regression model for mortality using those features - and the outcome would be comparable as well. ↵↵

Comments on the manuscript:

I think the authors have showed efforts to reflect reviewers' comments. I have just one comment to add - whether authors have thought about generating an implementable format of the prediction system. Graphic user interface could be one of them. It would be wonderful if there is a visual example of the Transformer model at the users' end. What would be the final outcome of the model performance to clinicians at bedside? In what format do authors anticipate the model to be delivered to bedside?

Comments on the supplementary materials:

Page 5 - Wondering whether following sentence is correct - ' Out of measurement range artifact. Laboratory values can be within the plausible range but exceeding the range of the measurement device.' Usually the range of the measurement device is much larger than the plausible range, since the machine usually have large enough coverage going outside the physiologic boundary. For example, extremely high blood pressure SBP > 400 cannot be measured by usual device. However, we generally think SBP greater than 300 to be implausible. Just wondering what the artifact removal steps/sequences were.

Version 2:

Reviewer comments:

Reviewer #4

(Remarks to the Author)

Re: 2nd rebuttal

Title: Machine Learning For Early Dynamic Prediction Of Functional Outcome After Stroke

I thank you for the authors' efforts to answer/address all questions and adding the revised text and data to the supplementation and main body of the manuscript. I believe readers will be able to learn key messages of the article with your endeavor.

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Early dynamic prediction of functional outcome after stroke using machine learning

Response to the reviewers:

We thank the reviewers for their comments about our manuscript. In the point-by-point response below, we tried to address the different issues raised. The article has been modified accordingly in the revised version. Thanks to the comments and suggestions of the reviewers, we hope that the manuscript has improved.

Reviewer #1:

Summary

The study presents a series of non-linear machine learning models that were developed and validated to predict the mortality and functional status of patients suffering from ischemic stroke within three months of their original hospital admission. The models integrate a variety of clinical, physiological, and biological data collected within the first 72 hours of hospital admission. The authors leverage data from the Geneva University Hospital from 2018 to 2021 to derive the models and the MIMIC-III database to validate them externally. A comparison of the discriminatory performance of a transformer-based neural network architecture, a recurrent neural network architecture, and the XGBoost algorithm is illustrated in the manuscript. The Shapley Additive Plots (SHAP) were applied to identify the key predictors driving each patient's outcome.

Comments

The study illustrates the application of a series of well-established machine-learning techniques to two mid-term stroke patient outcomes after hospital admission. I would like to commend the authors for using two different patient cohorts from different countries to ensure the generalizability of the proposed models. An additional strength of the paper is the diverse types of data that are used to characterize a patient's condition and the incorporation of both time-variant and time-invariant information. Nevertheless, there are various areas of improvement for the manuscript that I outline below.

Response to reviewer 1:

We thank Reviewer #1 for the thorough analysis of our work and for the suggestions for improvement. In the section below, we address each comment in a point-by-point response.

1. In its current form, the exposition of the manuscript does not meet the standards of the journal. I urge the authors to review the formatting and notation used across the manuscript to ensure that they conform to the international guidelines. In addition, the language used across the paper could be formalized, using more specialized vocabulary that is appropriate for a medical publication (see the second sentence of the Introduction).

Response to comment 1:

Based on the comment by reviewer #1, we have checked the wording in the new version of the manuscript to ensure conformity with international guidelines and removed unspecific formulations. We have also checked that our writing complied with the writing style of the *Communication Medicine* journal (<https://www.nature.com/documents/commsj-med-style-formatting-guide-accept.pdf>).

We thank the reviewer for giving us the opportunity to explain our approach including our writing style. Our aim was to reach out to a broad range of readers including clinicians, as we think our work may have important clinical relevance, even beyond the scope of stroke, but also readers interested in more technical aspects of machine learning applied to the clinical setting. Indeed, we developed advanced methods with the presentation and comparison of novel machine learning algorithms. To highlight both clinical and technical aspects of our work, we submitted our research to *Communication Medicine*. We used a language that we hope will appeal to the readers of the journal.

Comment 2. One of my greatest concerns relates to the definition of the outcome of interest. Most of the literature focusing on three-month mortality after an acute ischemic stroke sets the timepoint of data collection at the time of hospital discharge. On the contrary, most researchers leveraging time-varying data throughout the hospital admission recognize in-hospital outcomes, such as in-hospital mortality and morbidities, length of stay, and hospital readmission, as the priority endpoints that affect their decision-making process. Why do the authors focus on the outcomes of functional status and mortality in a three-month horizon, leveraging up to 72 hours of information instead of at the time of discharge? As long as the patient remains at the hospital, it is inevitable that the patient's status will change, and thus, the specific prediction after 72 hours becomes irrelevant. According to the literature review, the median and average length of stay after an acute ischemic stroke is around seven to ten days (see [1-2]), which raises doubts regarding the clinical relevance of these models.

Response to comment 2:

We thank the reviewer for his comments about the outcome measures that are relevant in stroke studies, particularly after ischemic events. Based on Comments 2 and 3 (reviewer #1), we have now used three outcome measures at two distinct timepoints, i) mortality at discharge, as well as ii) mortality and iii) morbidity at three months (pages 6-7, 12-13). Both timepoints have distinct specificities. Regarding mortality at discharge, the time interval is relatively short-term and may vary depending on factors such as the severity of the stroke and the length of hospital stay. Mortality at discharge provides immediate feedback on the effectiveness of acute interventions and hospital care. On the other hand, assessing morbidity and mortality at three months provides insights into the long-term outcomes and prognosis of stroke survivors, including their ability to regain independence and participate in daily activities.

Regarding our inclusion of data recorded only within the first 72 hours after admission, we chose this time interval because this period is the most critical, during which patients have the highest risk of early neurological deterioration¹. Therefore, this early time period is a turning point determining long-term clinical outcome. Consequently, during this time period, patients are monitored in an acute care ward in our centre, as well as in comparable international cohorts^{2,3} reflecting international recommendations⁴. This further allows comparison with the external validation cohort, as the MIMIC-III dataset is limited to patients in the ICU, with continuous monitoring. Prior studies in general ICU populations reported encouraging results using a 72 hour timeframe⁵. Finally, interventions targeting early stroke reoccurrence and early clinical deterioration, such as blood pressure and glucose control focus on the first 24 to 72 hours after the index event⁶⁻⁸. Integrating all data from the hospital stay into our model would increase the number of observations per admission 3 to 4-fold. Although a continuous predictive monitoring system arching over the entire hospital stay would be enviable, we have chosen to focus on the most critical time period, during which patients are closely monitored and most interventions occur, as a first step to balance benefit with computational complexity and cost^{9,10}.

This reflexion on data collection and outcome measures has been added to the limitations of the manuscript (see also response to comment 3, reviewer #1, as well as revised manuscript page 19).

Comment 3. I would recommend to the authors to alter their focus, following one of the two proposed directions outlined below:

a. Predict short-term outcomes like 3-month mortality or functional status at the time of discharge using varying time horizons of observation for each patient depending on the length of stay for their index admission.

b. Predict in-hospital mortality and morbidity outcomes using the 72-hour time frame that they have already established in the manuscript by changing the dependent variables in the model.

Response to comment 3:

We thank reviewer #1 for the suggestions. As recommended, we have added a third outcome measure: *in-hospital mortality*. Predictions were computed with the new outcome. We report the training procedure and the results for internal and external validation datasets in the updated manuscript as previously discussed (see response to Comment 2, as well revised manuscript pages 6-7, 12-13).

We nonetheless believe in the importance of also predicting 3-month outcomes using the 72-hour time frame. Indeed, 3-month clinical status is the most frequently used outcome for acute ischemic stroke interventions¹¹⁻¹⁴. Three-month modified Rankin scale (mRS) has become the main clinical measure used to determine the efficacy of interventions after acute ischemic stroke in randomized clinical trials, as it correlates well with long-term death and

disability¹⁵. The outcome at discharge is less frequently considered in clinical trials. However, we agree with reviewer #1 that adding an outcome measure such as mortality at discharge is of clinical relevance. The resulting decomposition into alive and well, alive and disabled or dead answers a common question for the patient himself, his family, and health-care personnel. We have added this reflexion to the discussion of the manuscript (page 19).

4. Another issue with the selected outcomes is that there is limited value in a flexible model that is adaptive to hourly information. Given that the endpoint lies in a time horizon that is significantly distant compared to the input information, the ability of the model to incorporate a flexible timeframe is not reflected in its application. This observation reinforces the need to choose a different scope (as described in the previous comment).

Response to comment 4:

As discussed in response to comments 2 and 3, we believe that the first 72 hours following an acute ischemic stroke are a critical time window for interventions aiming to improve medium to long-term outcomes. Moreover, it is the period at most risk for deterioration. For these reasons, patients are hospitalized in acute care wards during this timeframe, with continuous or high-frequency monitoring. A dynamic model, producing hourly updated predictions during this early period is of particular clinical value as it may signal patients at risk of bad long-term outcome to the care team, which, given adequate resources, can respond in a timely fashion.

Although improving long-term outcome is the ultimate goal for all interventions including those implemented in the acute phase, we fully acknowledge the importance of evaluating the impact of medical interventions within a shorter timeframe. As mentioned above, we have therefore added the prediction of in-hospital mortality to include an outcome within a shorter time horizon (pages 6-7, 12-13).

Comment 5. The literature review is incomplete as there have been other approaches that use time-series ICU data to predict short-term stroke mortality with transformer-based architectures [3-4]. The articles referenced above were retrieved after a quick search in online libraries. I recommend a more thorough review of the existing medical literature, including pre-trained models that use a transformer-based architecture that are tailored to the task of mortality prediction after an acute ischemic stroke, even with different types of data.

Response to comment 5:

We thank the reviewer for the opportunity to add a more extensive scope of the literature to the introduction. In addition to the afore-mentioned references, we introduce the reader to the applications of transformer models to the analysis of medical text, images, and event sequences in the setting of acute ischemic stroke (pages 4-5)¹⁶⁻²².

Comment 6. How did the authors filter the population to ensure that it involves only patients with acute ischemic stroke? As highlighted in [5], ICD codes are notoriously inaccurate in

capturing the true clinical condition of the patient. The specific study found that more than 30% of the patients reported in insurance claims with acute ischemic stroke did not actually have the condition. I would suggest conducting an independent audit of a random sample of the patients included in the study with expert physicians who could verify the patients' diagnosis.

Response to comment 6:

The diagnosis of ischemic stroke was established prospectively by a board-certified stroke neurologist. In the Geneva dataset, as part of clinical management, a discharge letter including the diagnosis is established for all patients and signed by a senior stroke neurologist. The patient data are then prospectively entered into the Swiss Stroke Registry. Auditing of this registry is regularly performed internally and by external authorities mandated by Swiss Health authorities. Magnetic resonance imaging was used as part of standard practice to identify stroke lesions. All patients included from the MIMIC-III dataset underwent a two-step verification process: 1) automated inclusion by ICD code and admission diagnosis and then 2) verification of inclusion and exclusion criteria based on the discharge note by an expert physician.

Respective method sections have been updated to better describe inclusion criteria (pages 6 & 10).

Comment 7. The study period involves some of the initial phases of the COVID-19 pandemic. Do the models account for whether a patient's stroke was provoked by COVID-19 or not? Can the authors provide any insights regarding the impact of COVID-19 on the mortality risk and functional status of patients who experience acute ischemic strokes?

Response to comment 7:

We did not account for a potential COVID-19 infection as putative causal agent in the proposed models. In a recent work from the centres participating in the Swiss Stroke Registry, we have shown that this is exceedingly rare in our population (8 / 2341)²³. Furthermore, establishing the causal link between the infection and the ischemic event remains controversial. These limitations make stroke provoked by COVID19 unlikely to be a good feature.

In the same work, we have however shown that a concomitant COVID19 infection adversely affects long-term outcomes. We have therefore added a subgroup analysis based on SARS-CoV-2 infection status during the stroke admission. Although this analysis is limited by the low number of events, we show that the ability to predict 3-month functional outcomes is preserved in COVID19 patients (page 9, Supplemental Tables 5-7).

Comment 8. There is a significant risk of cherry-picking the models when there is no bootstrapping with multiple random splits of the data between training/validation and testing sets in the derivation cohort. While the authors conduct bootstrapping the cross-

validation procedure, it is important to apply an equivalent process for the training/testing set split and report confidence intervals around AUC, accuracy, specificity, etc.

Response to comment 8:

We agree with the reviewer that there is significant risk of overfitting and cherry-picking when using advanced machine learning models. Consequently, we have taken great care to respect a thorough data separation and cross-validation scheme. We used 5-fold cross-validation to obtain training/validation splits used for parameter finetuning and model selection. A holdout test set was set apart at the beginning of the study and only used once for the final testing (internal validation) of the selected models, using bootstrapping to obtain 95% confidence intervals. Finally, we have used a different dataset for external validation (MIMIC-III) without any retraining of the obtained models. All metrics are reported with confidence intervals. We have clarified this procedure in the methods section and added a Supplemental Figure 2 for a visual representation of the data splitting (page 7, Supplemental Figure 2).

Comment 9. It is important for the readers to understand better the hyperparameters and specific architectures used for the neural network models. What kind of structure did the authors use? Did it involve a pre-trained NN? How many observations were generally used per admission?

Response to comment 9:

The transformer model is composed of a linear input embedding layer, followed by a positional encoding layer and a variable number of encoder layers with a multi-head self-attention mechanism, and a position-wise fully connected feed-forward network. The last layer of the network is a linear projection used for classification. No pretraining was used in any of the models. A mean of 857 observations per admission were used.

We now report with greater details the architecture and hyperparameters of the developed models in the Supplementary Methods 3.

Comment 10. The fact that some patients were associated with multiple admissions might positively bias the performance of the model. I recommend including only the first admission per patient.

Response to comment 10:

We thank the reviewer for his comment. Readmissions were only included in the training/validation splits of the cross-validation procedure and were always assigned to the same split. There were no readmissions in the holdout test set and external validation set in which model performance was evaluated. Therefore, the reported performance should not be affected. We have now clarified this aspect in the methods (page 7).

Comment 11. When delving into the results, the XGBoost model has equivalent and, in some

cases, even better performance compared to the transformer-based model. The XGBoost model, though, does not have the ability to incorporate varying time-series information. How did the authors train this model? What was the structure of the feature vector that contained the time-series information? Why should clinicians prefer a more complex model that would be harder to implement in practice compared to the ensemble learner?

Response to comment 11:

As the XGBoost model does not inherently handle timeseries, we aggregated each feature into current value, mean, minimum and maximum up to a given timestep. Therefore, the input to the model was a vector of length 4*(number of features). We have detailed the input space and training paradigm for each model in the Supplemental Methods 3 & 4. The XGBoost model performs similarly in most domains, but with a worse Precision/Recall trade-off than the transformer model. Furthermore, we believe that the transformer architecture will find increasing use in the processing of multimodal medical data and that this work shows its potential in the processing of timeseries. Nonetheless, we agree that there will be use-cases in which a more lightweight model will be preferred even at the cost of slightly lower performance. For this reason, the overall proposed framework is backbone agnostic and we open-sourced the weights of all developed models. The clinician using the framework can thus opt for the model of his choice according to his needs. This reflection has been added to the discussion (page 17).

Comment 12. The authors could add a dedicated section in the Supplemental Material with more information regarding the mRS scale and the implications of each status.

Response to comment 12:

We have now added Supplemental Table 9 detailing the modified Rankin scale and its clinical implications.

References provided by Reviewer #1:

1. Chang, Ku-Chou, et al. "Prediction of length of stay of first-ever ischemic stroke." *Stroke* 33.11 (2002): 2670-2674.
2. Huang, Ying-Chih, et al. "The impact factors on the cost and length of stay among acute ischemic stroke." *Journal of Stroke and Cerebrovascular Diseases* 22.7 (2013): e152-e158.
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processing methods to identify ischemic stroke, acuity and location from radiology reports." PloS one 15.6 (2020): e0234908.

Reviewer #2:

This manuscript addresses the prediction of functional outcome of stroke patients based on continuously acquired clinical data points. The methodology and experimental framework are well described, and the authors are doing an excellent job at introducing the landscape of stroke and the uncertainty associated with its treatment.

In particular, the evaluation of the framework is well designed as the models are trained on a large cohort of patients from the authors' institution and evaluated on a publicly available dataset with similar variables. This dramatically boosts the readers' interest as it is a good indication of the generalization power of the models introduced in the study.

Response to reviewer 2:

We thank the reviewer for the comments and review, allowing us to significantly improve the original work.

Comment 1. While I did not notice significant weaknesses, a few concepts are a bit vague and should be more precisely described:

- Three machine learning methods were used and compared. While LSTM can take into account temporal sequences, it is not clear how the other two methods (XGboost and Transformer) were being used on continuous data. Was the input the 1-hour summary/average? How about for LSTM - was it a sliding window? If the data of each hour is inputted independently to the model, what patterns could be missing? Is there any temporal trend that could be captured with the current approach?

Response to comment 1:

Continuous features were downsampled into hourly median, minimum and maximum values. The resulting timeseries was then used in its entirety as input for the LSTM and transformer model with a shape of (number of timesteps) x (number of features). Each hourly input is thus not independent to the previous timesteps allowing even complex temporal trends to be easily captured by the models.

As the XGBoost model does not inherently handle timeseries, we aggregated each feature into current value, mean, minimum and maximum up to a given timestep. Therefore, the input to the XGBoost model was a vector of length 4*(number of features). Although this limits its ability to capture more complex temporal trends, such as recovery from local minima, global trends can still be interpreted by the model by comparing the current value to the computed mean/min/max.

We have now detailed the input space of each model in the Supplemental Methods 4.

Comment 2: - ROC and PR curves are reported in figure 2 and represent the performance of the models on the MIMIC III dataset. What were the results on the training dataset after cross-validation?

Response to comment 2:

We have now added ROC and PR curves on the training and validation splits as Supplemental Figures 4 & 5. These should however be interpreted with care as there is a very high risk of overfitting.

Comment 3: - As mentioned by the authors, imaging findings were left out of the model. There is a great opportunity to evaluate them within the framework as it would provide an indication of their predictive power (even if only done on subset of patients from the local cohort). While using raw imaging is challenging, it seems very doable to incorporate findings from the neuro-radiology report such as volume of the core, penumbra, bleeding, and other commonly reported features. This would greatly improve the originality and the impact of the study.

Response to comment 3:

We thank the reviewer for his recommendation and we have now evaluated the predictive power of imaging features extracted from the analysis of the admission perfusion imaging CT (volumes of Tmax > 4/6/8/10s, and CBF < 30%) for a random subset of the Geneva Stroke Dataset. These features were used as additional input by the transformer model, when available. In the holdout test set, we have evaluated the performance of this imaging model in patients for whom imaging features had been extracted, once with access to imaging data, and once hiding the imaging data from the model. We report slightly higher performance for the model having access to imaging data. This analysis has been added to the methods, results, and discussion sections (pages 10, 14 and 18). We further added Supplemental Tables 2 & 8.

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Machine Learning For Early Dynamic Prediction Of Functional Outcome After Stroke

Response to the reviewers:

We thank the reviewers for their comments about our revised manuscript. In the point-by-point response below, we tried to address the different issues raised. The article has been modified accordingly in the updated version. Thanks to the comments and suggestions of the reviewers, we hope that the manuscript has improved. We further adapted the title of the manuscript to better reflect the contents of our study.

Reviewer #3:

The authors have satisfactorily addressed previous reviewer concerns. They have included additional clarifying details and context that improves the quality of the manuscript.

Response to reviewer 3:

We thank Reviewer #3 for the comment regarding the revised version of the manuscript.

Reviewer #4:

Summary

Thanks for the opportunity in reviewing the original research work written by Klug et al, titled as 'Early dynamic prediction of functional outcome after stroke using machine learning'. This work has been submitted, recommended to be revised, and being reviewed again for its appropriateness and completeness for journal publication.

Response to reviewer 4:

We thank Reviewer #4 for the analysis of our work and for the suggestions for improvement. In the section below, we address each comment in a point-by-point response.

Comments

1. *I found the article to be very intriguing since there are not a lot of literature suggesting the ML models' role in addressing functional outcome after acute stroke. However at the same time, ironically, the results authors presented also showed the characteristic 'prediction was predictable' tendency that comes with many ML models that sometimes nullify its value. performance of the Transformer (encoding) model was not drastically better than the XGBoost, and raised question on the absolute superiority of the sophisticated ML model with high-level inference capacity on this type of prediction. For example, functional outcome (authors used modified Rankin) could be all very predictable if NIHSS, pre-stroke disability, or IVT timing (which SHAP showed) were given to general neurologists in clinical setting without using ML models in their practice. I think the authors could probably construct a logistic*

regression model for mortality using those features - and the outcome would be comparable as well.

Response to comment 1:

We thank the reviewer for this comment. We fully agree that the simplest model will be preferred in clinical practice. As suggested by the reviewer, we have evaluated the prediction of functional and survival outcomes using the variables suggested by the reviewer (NIHSS, pre-stroke disability, and IVT timing). The AUC for the prediction of functional outcome was 0.799 (0.748-0.840). The performance of this simplified model was acceptable, but inferior to the more sophisticated machine learning models. This analysis was added to the compared models (Methods, p. 9; Supplemental Table 3, p. 15-17).

We agree with the reviewer that prediction is highly dependent on prehospital handicap and clinical severity on admission. Although not evaluated here, clinician performance for the prediction of functional outcome was assessed in a recent study by Herzog et al, reporting an accuracy of 64% [59.8%–68.4%] in a selected population with large vessel occlusion¹. The added benefit of the proposed model is however mainly the dynamic and continuous nature of the prediction. At every timepoint, a change in trajectory as well as variables impacting outcome can be identified. Further studies are however needed to evaluate whether acting upon variables associated with a predicted deterioration would impact outcomes favorably.

2. Comments on the manuscript: I think the authors have showed efforts to reflect reviewers' comments. I have just one comment to add - whether authors have thought about generating an implementable format of the prediction system. Graphic user interface could be one of them. It would be wonderful if there is a visual example of the Transformer model at the users' end. What would be the final outcome of the model performance to clinicians at bedside? In what format do authors anticipate the model to be delivered to bedside?

Response to comment 2:

The primary goal of the developed system was to serve as proof of concept for a backbone of a continuous monitoring tool assisting the treating physician. As suggested, we have developed a demo version of a possible graphical user interface giving access to the main features of the model. The user can see the predicted probabilities for functional outcome and mortality in real-time for a given patient, along with their contributing factors. Values from the past hours are plotted and changes in predicted outcome are visually marked. Alerts could be triggered by an increase in predicted mortality by over 10% or an absolute predicted mortality in the 90th percentile². A screenshot of the graphical user interface has been added as Supplemental Figure 7. The desired impact of the model is to assist the clinician by providing as accurate as possible predictions for discussions with the treating team and families, to help detect deteriorations earlier and point towards contributing factors.

3. Comments on the supplementary materials: Page 5 - Wondering whether following sentence is correct - ' Out of measurement range artifact. Laboratory values can be within the

plausible range but exceeding the range of the measurement device.' Usually the range of the measurement device is much larger than the plausible range, since the machine usually have large enough coverage going outside the physiologic boundary. For example, extremely high blood pressure SBP > 400 cannot be measured by usual device. However, we generally think SBP greater than 300 to be implausible. Just wondering what the artifact removal steps/sequences were.

Response to comment 3:

We thank the reviewer for the opportunity to clarify our preprocessing. An illustrative example of an “out of measurement range artifact” would be the reading “High” on a capillary glucometer. Although the measurement range is capped at 30 to 80 mmol/l depending on the manufacturer, the real value can be plausibly above this maximum. For a glucometer with a maximum reading of 60 mmol/l, the value would have been set to 63 mmol/l. Finally, values outside of the plausible range 0.5-87 mmol/l were set as missing. This example detailing the artifact removal steps was added to the Supplementary Methods 1 (p. 5).

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